Lesson plan · for the teacher

Microcontrollers: sense, decide, act

IoT · Level 2 · Lesson 4

Time
40 min

Materials
A torch, and a book or curtain to shade the window; Program cards on paper: read, decide, act, wait; Optional: a phone or computer with a browser, for MakeCode or Wokwi; Optional: a micro:bit, or an Arduino with an LED, a 220 Ω resistor and a light sensor

Learning goals

- Name the parts of a microcontroller board: power, pins, inputs and outputs.
- Explain how a program's loop reads a sensor, decides and acts (a night light).
- Choose a way to start: a simulator, a real board, or unplugged.

Activity: Be the microcontroller

1. Warm-up (5 min): Ask pupils for devices that switch themselves on or off: street lights, automatic doors, a fridge, a phone screen. List what each one senses.
2. Teach (8 min): Draw a board with inputs on one side and outputs on the other. Write the loop as four boxes: read, decide, act, wait. Write the night-light rule: “dark below 300”.
3. Act it out (10 min): One pupil holds the program card and a torch and stands by the window. Classmates shade the window with a book; she reads, decides and switches the torch, then waits five seconds and repeats.
4. Flowchart (12 min): Groups draw the loop as a flowchart. Then change the threshold and predict: will the light come on earlier or later in the evening?
5. Wrap-up (5 min): If you have a phone or computer, show the night light in MakeCode's micro:bit simulator and move the light slider. Exit ticket: the loop in four boxes.

Write the program card before class: READ the window light; DECIDE: is it darker than our level?; ACT: torch on or off; WAIT 5 seconds; REPEAT. Try MakeCode on your own phone first: the micro:bit's light level runs from 0 to 255, so use a threshold such as 50.

Discussion

- Which devices around you run a loop like this?
- Why is it useful that the threshold is a number we can change?
- What could go wrong if the light sensor sits in the shade?

Exit ticket

- Draw the night light's loop in four boxes, with the rule in the decide box.

Try it at school tomorrow

Be the microcontroller: One pupil plays the microcontroller with a program card: read the light at the window, decide with the rule (dark below a level the class agrees), and switch a torch on or off. Classmates change the light with a book or a curtain. Then groups draw the program as a flowchart and change the threshold. If you have a phone or computer, build the same night light in a simulator.

Practice questions · for you

Microcontrollers: sense, decide, act

IoT · Level 2 · Lesson 4

1 **On cloudy days, the corridor night light switches on at noon. What should you change?**

- ◯ Make the loop run faster, so it checks more often
- ◯ Raise the threshold, say from 300 to 400
- ◯ Lower the threshold, say from 300 to 200

2 **Match each part to what it does.**

A button	• • • •	Analog input: a number from 0 to 1023
A light sensor	• • • •	Output: light, through its resistor
An LED	• • • •	Digital input: on or off
A water pump	• • • •	Runs once, at power-on
Setup	• • • •	Output through a driver or relay

3 **Put the night light's program in order.**

- ☐ Read the light sensor
- ☐ Power on: setup runs once
- ☐ Act: switch the LED on or off
- ☐ Wait a moment, then read again
- ☐ Decide: is the reading below 300?

4 **You can connect a water pump straight to a microcontroller's output pin, as long as the program is right.**

- ◯ True
- ◯ False

5 **Your school has no boards yet. Which let pupils still write and test a night-light program?**

- ☐ The free Wokwi Arduino simulator
- ☐ A pupil playing the microcontroller with a program card
- ☐ The free MakeCode micro:bit simulator in a browser
- ☐ Waiting until the school can buy kits
- ☐ Copying the code into notebooks without testing it

✎ **Which device at your school or in your town has a microcontroller inside? What does its loop read, decide and do?**

Check your answers

Microcontrollers: sense, decide, act

IoT · Level 2 · Lesson 4

1 **Lower the threshold, say from 300 to 200**

Dark means below the threshold, so a lower threshold means only a really dark corridor counts as dark.

2 **A button — Digital input: on or off**
A light sensor — Analog input: a number from 0 to 1023
An LED — Output: light, through its resistor
A water pump — Output through a driver or relay
Setup — Runs once, at power-on

Inputs bring news in; outputs act. Anything that needs more than a pin's small current, like a pump, goes through a driver or relay.

3 **Top to bottom: 2, 1, 4, 5, 3**

Setup comes once, then the loop repeats: read, decide, act, wait.

4 **False**

A pin gives only about 20 mA, far less than a pump needs, and the pump's power can damage the board.

5 **The free MakeCode micro:bit simulator in a browser; The free Wokwi Arduino simulator; A pupil playing the microcontroller with a program card**

Simulators run on a phone or computer for free, and acting it out tests the logic with no device at all.